

1

Thermostatics

The goal of this first chapter is to give a presentation of thermodynamics, due to H. Callen, which will allow us to make the most direct connection with the statistical approach of the following chapter. Instead of introducing entropy by starting with the second law, for example with the Kelvin statement ‘there exists no transformation whose sole effect is to extract a quantity of heat from a reservoir and convert it entirely to work’, Callen assumes, in principle, the existence of an entropy function and its fundamental property: the principle of maximum entropy. Such a presentation leads to a concise discussion of the foundations of thermodynamics (at the cost of some abstraction) and has the advantage of allowing direct comparison with the statistical entropy that we shall introduce in Chapter 2. Clearly, it is not possible in one chapter to give an exhaustive account of thermodynamics; the reader is, instead, referred to classic books on the subject for further details.

1.1 Thermodynamic equilibrium

1.1.1 Microscopic and macroscopic descriptions

The aim of statistical thermodynamics is to describe the behaviour of macroscopic systems containing of the order of $N \approx 10^{23}$ particles.¹ An example of such a macroscopic system is a mole of gas in a container under standard conditions of temperature and pressure.² This gas has 6×10^{23} molecules³ in incessant motion, continually colliding with each other and with the walls of the container. To a first approximation, which will be justified in Chapter 2, we may consider these molecules as classical objects. One can, therefore, ask the usual question of classical mechanics: given the initial positions and velocities (or momenta) of the

¹ With some precautions, one can apply thermodynamics to mesoscopic systems, i.e. intermediate between micro- and macroscopic, for example system size of the order of $1 \mu\text{m}$.

² The reader will allow us to talk about temperature and pressure even though these concepts will not be defined until later. For the moment intuitive notions of these concepts are sufficient.

³ In the case of a gas, we use the term ‘molecules’ instead of the generic term ‘particles’.

molecules at $t = 0$, what will the subsequent evolution of this gas be as a function of time? Let us, for example, imagine that the initial density is non-uniform and ask how the gas will evolve to re-establish equilibrium where the density is uniform. Knowing the forces among the molecules and between molecules and walls, it should be possible to solve Newton's equations and follow the temporal evolution of the positions, $\vec{r}_i(t)$, $i = 1, \dots, N$, and momenta, $\vec{p}_i(t)$, as functions of the positions and momenta at $t = 0$. We could, therefore, deduce from the trajectories the evolution of the density $n(\vec{r}, t)$. Even though such a strategy is possible in principle, it is easy to see that it is bound to fail: if we simply wanted to print the initial coordinates, at the rate of one coordinate per microsecond, the time needed would be of the order of the age of the universe! As for the numerical solution of the equations of motion, it is far beyond the capabilities of the fastest computers we can imagine, even in the distant future. This kind of calculation, called molecular dynamics, can currently be performed for a maximum of a few million particles.

The quantum problem is even more hopeless: the solution of the Schrödinger equation is several orders of magnitude more complex than that of the corresponding classical problem. We keep in mind, however, that our system is, at least in principle, susceptible to a microscopic description: positions and momenta of particles in classical mechanics, their wave function in the quantum case. If this information is available, we will say that a system has been attributed a *microscopic configuration* or *microstate*. In fact, this microscopic description is too detailed. For example, if we are interested, as above, in the temporal evolution of the density of the gas, $n(\vec{r}, t)$, we have to define this density by considering a small volume, ΔV , around the point $\vec{r}$, and count (at least in principle!) the average number of gas molecules in this volume during a time interval, Δt , centred at t . Even though ΔV is microscopic, say of the order of $1 \mu\text{m}$ on a side, the average number of molecules will be of the order of 10^7 . We are only interested in the average number of molecules in ΔV , not in the individual motion of each molecule. In a macroscopic description, we need to make spatial and temporal averages over length and time scales that are much larger than typical microscopic scales. Length and time scales of $1 \mu\text{m}$ and $1 \mu\text{s}$ are to be compared with characteristic microscopic scales of 0.1 nm and 1 fs for an atom. In this averaging process, only a small number of combinations of microscopic coordinates will play a rôle, and not each of these coordinates individually. For example, we have seen that to calculate the density, $n(\vec{r}, t)$, we have to count all molecules found at time t in the volume ΔV around point $\vec{r}$, or, mathematically,

$$n(\vec{r}, t) = \frac{1}{\Delta V} \int_{\Delta V} d^3r \sum_{i=1}^N \delta(\vec{r} - \vec{r}_i(t)) \quad (1.1)$$

This equation selects a particular combination of positions, $\vec{r}_i(t)$, and gives an example of what we will call *macroscopic variables*. Another example of a combination of microscopic coordinates yielding a macroscopic variable will be given below for the energy (Equation (1.2)).

Since the microscopic approach leads to a dead end, we change descriptions, and take as fundamental quantities global macroscopic variables related to the sample: number of molecules, energy, electric or magnetic dipole moment, etc. Macroscopic variables, or more precisely, their densities (density of molecules, of energy etc.) define a *macrostate*. The evolution of macroscopic variables is governed by deterministic equations: Newton's equations for elastic objects, Euler's equations for fluids, Maxwell's equations for electric or magnetic media, etc. However, this purely mechanical description is insufficient since a macrostate is compatible with a very large number of different microstates. Therefore, we cannot forget the microscopic degrees of freedom that have been eliminated by averaging. For these microscopic degrees of freedom, which, for the moment, we have ignored in the macroscopic approach, we will use a probabilistic description; this will in turn lead to the concept of entropy, which is needed to complete our macroscopic picture. This probabilistic approach will be discussed in Chapter 2. Contrary to other macroscopic variables, *entropy is not a combination of microscopic variables*: it plays a singular rôle compared to other macroscopic quantities.

In the remainder of this chapter, we will limit ourselves to a thermodynamic description and only consider macroscopic variables and the entropy.

1.1.2 Walls

A particularly important macroscopic variable in thermodynamics is the energy, which can take many forms. In a mechanical system with only conservative forces (derivable from a potential), the mechanical energy, which is the sum of the kinetic and potential energies, is conserved. A mechanical system protected from all external influences, i.e. *isolated*, finds its energy conserved, in other words independent of time. Mathematically, the energy can be written as

$$E = \sum_{i=1}^N \frac{\vec{p}_i^2}{2m} + \frac{1}{2} \sum_{i \neq j} U(\vec{r}_i - \vec{r}_j) \quad (1.2)$$

To simplify the writing, we have assumed in this equation that the particles are identical and of mass m ; $\vec{p}_i$ is the momentum of particle i , $\vec{r}_i$ its position, and U the potential energy of two molecules. We also assumed the molecules to have no internal structure. Equation (1.2) also gives the expression for the classical or quantum Hamiltonian, H , of the isolated system. In the quantum case, $\vec{p}_i$ and

$\vec{r}_i$ are the canonically conjugate momentum and position operators for particle i . When the system is not isolated, we know that we can transfer mechanical energy to it: when we compress a spring, it acquires additional energy, which it stores in the form of elastic potential energy. During the compression process, the point where the force is applied moves with the consequence that energy is given to the system in the form of *work*. Similarly, we supply energy to a gas by compressing it with the help of a piston. In both cases, the *external parameters*⁴ of the system, length of the spring in one case, volume of the gas in the other, are modified in a known way. However, we know, experimentally, that we can transfer energy to an object in many other ways. Any handyman knows that we can transfer energy to a drill bit by drilling a hole in concrete. The bit heats up due to friction, and, according to a popular but thermodynamically incorrect statement (Footnote 10), some of the mechanical energy supplied by the drill motor is 'changed to *heat*'. We can obtain the same result by leaving the drill bit in the sun on a hot summer day, which corresponds to 'transforming electromagnetic energy into heat', or by immersing it in boiling water, i.e. by using thermal contact. In the latter case, there is no visible modification of external variables (see Section 2.5.1) either of the bit or of the water. Only the microscopic degrees of freedom are involved in the exchange of energy. The heating of the bit corresponds to bigger vibrations of its atoms around their equilibrium positions, the concomitant cooling of the water corresponds to a reduction of the average speed of its molecules.⁵ *Energy transfer in the form of heat is characterized by the fact that neither the external parameters of the system, nor the configuration of the external medium, are modified.* This heat transfer can be effected by conduction (the drill bit in contact with water), or by radiation (between the sun and the bit).

In summary, a system can receive energy either in the form of work, or in the form of heat. The energy supplied in the form of work is, at least in principle, measurable from mechanical considerations because work is supplied by a macroscopic mechanical device whose parameters (masses, applied forces, etc.) are, supposedly, perfectly known.⁶ Work is obtained by causing a change, either of the external parameters, or the configuration of the external medium, or both. However, the amount of energy received by an object is not known with precision from the principle of conservation of total energy unless we are able to eliminate energy exchange in the form of heat. This can be accomplished by isolating the system using a heat insulating wall, or an *adiabatic wall*; on the other hand, a *diathermic wall*

⁴ External parameters are those quantities that are under the direct control of the experimentalist: volume, external electric or magnetic fields, etc.

⁵ To simplify the discussion, we neglect complications due to the potential energy of the molecules.

⁶ From this point of view, the energy supplied by an electric device will be considered as work since it can be determined by electric measurements performed with a voltmeter or an ammeter.

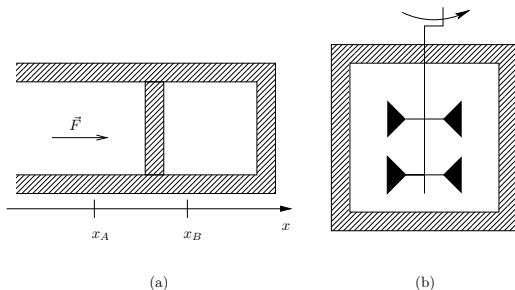


Figure 1.1 Two ways to supply work. (a) Compressing a gas, (b) Joule's experiment.

allows heat transfer. Inversely, we eliminate energy transfer in the form of work by using a *rigid wall* not penetrated by any mechanical device.⁷ The possibility of a thermodynamic description is founded on the existence, at least in theory (since walls are never perfectly adiabatic or rigid!), of walls capable of controlling the different forms of energy transfer.

An *isolated system* is a system that cannot exchange energy with its surroundings in any form whatsoever: it is isolated from the external world by walls that are perfectly adiabatic, rigid, impermeable to molecules and shielded electrically and magnetically.

1.1.3 Work, heat, internal energy

We now develop more quantitatively the concepts defined above. First we need to make an essential distinction between two different ways of supplying work. To fix ideas, let us consider work done on a gas by compressing it with the help of a piston displaced between positions x_A and x_B (Figure 1.1(a)), the position of the piston being given by its abscissa x . If at every instant the position of the piston and $F(x)$, the component of the applied force parallel to Ox , are perfectly controlled,

⁷ Rigorously speaking, for a mechanical system one also should eliminate energy transfer by other processes such as electric or magnetic.

the work $W_{A \rightarrow B}$ can be calculated by integrating between x_A and x_B the element of work δW

$$\begin{aligned}\delta W &= F(x) dx \\ W_{A \rightarrow B} &= \int_{x_A}^{x_B} dx F(x)\end{aligned}\tag{1.3}$$

While compressing the gas, the external parameter, x , and the force, $F(x)$, are entirely under the control of the experimenter who performs a *quasi-static transformation*. This important notion will be defined in general in Section 1.2.2. An example where work transfer is *not* quasi-static is studied in Problem 1.7.1. A vertical cylinder containing gas is closed with a piston on which a weight is placed. When this weight is suddenly removed, the gas expands and the piston reaches a new equilibrium position after a few oscillations. Neglecting the friction between the piston and the cylinder, the work supplied to the gas is $-P_{\text{ext}}\Delta V$, where P_{ext} is the external pressure and ΔV the change in volume. This change in volume was not controlled during the expansion of the gas. Another example of non-quasi-static transfer of work is illustrated in Figure 1.1(b). The system is isolated from the exterior by an adiabatic wall, but a motor turns vanes in the fluid, which heats up due to viscosity. The energy supplied in the form of work can be calculated from the characteristics of the motor.⁸ These two examples of non-quasi-static work transfer appear very different but do have a point in common. In the first example, the final temperature is higher than it would have been had the change been quasi-static, and, as is the case in the second example, viscous forces are responsible for the increase in temperature. The example of the heated drill bit, given earlier, is another illustration of work done non-quasi-statically.

We now examine the energy transfer between states of a system. Let A and B be two possible arbitrary states. The energy of each of these states is, in principle, a well-defined quantity, for example by Equation (1.2). In thermodynamics, this is called the *internal energy* and will be denoted by E . We know that only energy differences have physical meaning, and that *a priori* the interesting quantity is $E_B - E_A$. Our goal is to demonstrate that this energy difference is accessible experimentally. Note that all transferred energies, be they in the form of work or heat, are algebraic quantities that can be positive or negative.

Taking E_A as the reference energy, we will be able to determine E_B if it is possible to go from A to B by supplying the system only with work, positive or

⁸ A more modern version of this experiment, which dates back to Joule, consists of putting in the fluid a known resistance across which we apply a known potential difference: the amount of electrical energy 'transformed into heat' is thus known.

negative, since this work, furnished by a mechanical device, is measurable. To determine whether such a transformation is possible, we start from the following empirical observation. It is possible to go either from state A to state B , or from state B to state A by a process whose sole effect is to supply work to the system. However, under this condition, only one of the two transformations is allowed. We justify this statement as follows. If states A and B have the same volume, and if $E_B > E_A$, a mechanism similar to that in Figure 1.1(b) allows us to go from A to B , which will be impossible if $E_B < E_A$.⁹ If the volumes of A and B are different, we can use an adiabatic expansion or compression, $A \rightarrow A'$, which brings the system to the desired volume, $V_{A'} = V_B$, with an energy $E_{A'}$. If $E_{A'} < E_B$, work can be done to arrive at the final state with energy E_B . To summarize, we can determine, either $E_B - E_A$ by a transformation $A \rightarrow B$, or $E_A - E_B$ by a transformation $B \rightarrow A$, by supplying only measurable work to the system.

If the transformation $A \rightarrow B$ is now performed in an arbitrary manner, in other words it involves an exchange of both work and heat, we can control the work, $W_{A \rightarrow B}$, which is determined by macroscopic mechanical parameters. The energy difference, $(E_B - E_A)$, has previously been determined, and we thus obtain the amount of heat, $Q_{A \rightarrow B}$, supplied in this process

$$Q_{A \rightarrow B} = (E_B - E_A) - W_{A \rightarrow B} \quad (1.4)$$

This equation, which simply expresses conservation of energy, constitutes the 'first law of thermodynamics'. It is often written in the differential form

$$\delta Q = dE - \delta W \quad (1.5)$$

Unlike the increase in the internal energy, $E_{A \rightarrow B} = E_B - E_A$, the work, $W_{A \rightarrow B}$ and the amount of heat, $Q_{A \rightarrow B}$, are not determined by the initial and final states: they depend on the transformation itself.¹⁰ This is why, unlike the differential dE , the infinitesimal quantities δQ and δW are not differentials. We can understand this intuitively by making an analogy with mechanics. If a force, $\vec{F}$, is such that $\vec{\nabla} \times \vec{F} \neq 0$, the work it does between points A and B

$$W_{A \rightarrow B} = \int_A^B \vec{F} \cdot d\vec{l}$$

does not depend only on the points A and B . It also depends on the path taken, and there is no function whose differential gives the infinitesimal work.

⁹ Anticipating what is to follow, $E_B > E_A$ means that the temperature, T_B , of B is greater than T_A of A : it is impossible to cool down a volume simply by exchanging work.

¹⁰ We cannot, therefore, ascribe to a system a work or heat content. The concepts of heat and work expose two different forms of *energy exchange* between two systems: heat and work are 'energy in transit'.

1.1.4 Definition of thermal equilibrium

For simplicity, we limit ourselves to the ideal case of a system that is shielded from electric or magnetic influences, leaving these more complex cases to the references. This restriction is not a limitation of the theory, but merely a convenience to simplify the discussion. Suppose that our system is isolated in a completely arbitrary initial state, for example with a spatially dependent density. *Experience tells us that if we wait long enough, the system will evolve to an equilibrium state, that is a state which depends neither on time nor on the past history of the system.* The equilibrium state is entirely characterized by macroscopic variables and external parameters describing the system: the volume, V , the energy, E , and the numbers, $N^{(1)}, \dots, N^{(r)}$, of molecules of type 1, $\dots, r$.

The time that characterizes the approach to equilibrium is called the *relaxation time*. Relaxation times can be as short as a few microseconds, and as long as several millennia. It is, therefore, not obvious in practice to decide whether or not we have attained an equilibrium state. In numerous cases, we only reach a state of metastable equilibrium whose average lifetime can be extremely long. Such a state only appears to be independent of time, and in fact also depends on its past history. A very familiar example is hysteresis: if we magnetize an initially unmagnetized sample of magnetic material by applying a magnetic field, the magnetization does not disappear when the field is removed. The evolution of the magnetization, M , as a function of the applied magnetic field, B , describes a hysteresis cycle as shown in Figure 1.2. On this figure, the dashed line represents the case where the initial magnetization is zero. The magnetic state, therefore, depends on its past history even though we have obtained a magnet that is apparently in a stable state. However, we may, for example, reach a state whose magnetization is opposite to the magnetic

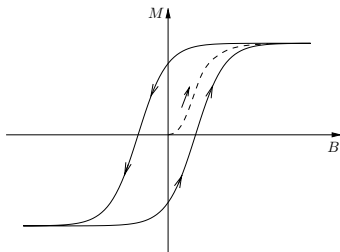


Figure 1.2 Hysteresis cycle for a magnetic system in an external field.

field, a state that is clearly metastable. Another common case is the existence of a variety of metastable crystals: for example, graphite is the stable crystalline form of carbon under standard conditions of pressure and temperature, and diamond is metastable. There are many other examples: glasses, alloys, memory materials, etc. In such cases, if we mistakenly assume that the system is at equilibrium, we may arrive at conclusions that contradict experiment.

While keeping in mind the difficulty related to extremely long relaxation times, we take as our first postulate the existence of equilibrium states: an isolated system will attain, after sufficiently long time, an equilibrium state that is independent of its past history and characterized by its own intrinsic properties such as volume V , energy E and numbers $N^{(i)}$, of molecules of different types. In what follows, we shall often limit ourselves to a single type of molecule, N in number, and to homogeneous equilibrium states whose properties, for example the density, are uniform. The quantities E , V , and N are said to be *extensive*: if we merge into one system two identical subsystems at equilibrium, the energy, volume and number of molecules are doubled.

1.2 Postulate of maximum entropy

1.2.1 Internal constraints

As we have emphasized in the introduction, our presentation of thermodynamics postulates the existence of an entropy function. To define it correctly, it is necessary to introduce the notion of *internal constraint*, of which we shall give a simple example. Consider an isolated system that we have divided in two subsystems (1) and (2), separated by a piston (Figure 1.3). As always, we assume that the contributions of the walls (or the piston) to the energy, E , to the volume, V , and to the total number of molecules, N , are negligible since they arise from surface effects. Consequently, E , V , and N represent the sums of energies, volumes and

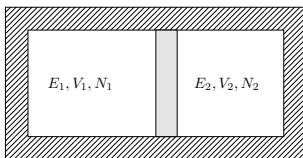


Figure 1.3 An isolated system which is divided into two 'subsystems'.

numbers of particles in the two subsystems

$$E = E_1 + E_2 \quad V = V_1 + V_2 \quad N = N_1 + N_2 \quad (1.6)$$

The piston creates the following internal constraints:

- If it is fixed, it creates a constraint in preventing the free flow of energy from one subsystem to the other in the form of work.
- If it is adiabatic, it creates a constraint in preventing the free flow of energy from one subsystem to the other in the form of heat.
- If it is impermeable to molecules, it creates a constraint in preventing the flow of molecules from one subsystem to the other.

We lift a constraint by rendering the piston mobile, diathermic or permeable. We can, of course, lift more than one constraint at a time.

Let us start with the following initial situation: the piston is fixed, adiabatic, impermeable, and both subsystems are separately at equilibrium. We lift one (or several) of the constraints and we await the establishment of a new equilibrium state. We can then pose the following question: what can we say about this new equilibrium state? We shall see that the answer to this fundamental question is given by the principle of maximum entropy.

1.2.2 Principle of maximum entropy

We make the following postulates, which are equivalent to the usual statement of the ‘second law of thermodynamics’:

- (i) *For any system at equilibrium, there exists a positive differentiable entropy function $S(E, V, N^{(1)}, \dots, N^{(r)})$.¹¹ As a general rule, this function is an increasing function of E for fixed V and $N^{(i)}$.¹²*
- (ii) *For a system made of M subsystems, S is additive, or extensive: the total entropy S_{tot} is the sum of the entropies of the subsystems,*

$$S_{\text{tot}} = \sum_{m=1}^M S(E_m, V_m, N_m^{(1)}, \dots, N_m^{(r)}) \quad (1.7)$$

- (iii) *Suppose the global isolated system is initially divided by internal constraints into subsystems that are separately at equilibrium: if we lift one (or more) constraint, the final entropy, after the re-establishment of equilibrium must be greater than or equal to the initial entropy. The new values of $(E_m, V_m, N_m^{(i)})$ are such that the entropy can only increase or stay unchanged. In summary: the entropy of an isolated system cannot decrease.*

¹¹ The reader will remark that S is, at the same time, a function of an external parameter V , and macroscopic variables E and N .

¹² See Problem 3.8.2 for an exception.

We emphasize that *a priori* entropy is only defined for a system at equilibrium. When a system is very far from equilibrium, we cannot in general define a unique entropy. However, when a system, not globally at equilibrium, can be divided into subsystems that are almost at equilibrium with their neighbours with whom they interact weakly, we can, once again, define the global entropy of the system with the help of Equation (1.7). As we shall see in more detail in Chapter 6, this permits us to define the entropy of a system that is not in global but only in local equilibrium. This case is very important in practice: for example, all of hydrodynamics rests on the notion of local equilibrium.

We now introduce the notion of *quasi-static processes*, already evoked in Section 1.1.2. A transformation $A \rightarrow B$ is said to be quasi-static if the system under consideration stays infinitesimally close to a state of equilibrium. A quasi-static transformation is necessarily infinitely slow since we need to wait after each step for equilibrium to be re-established. Clearly, a quasi-static transformation is an idealized case: a real transformation cannot be truly quasi-static. Along with walls that are perfectly adiabatic or perfectly rigid, quasi-static transformations are theoretical tools for formulating precise reasonings in thermodynamics. In practice, we consider a transformation quasi-static if it takes place over a time much larger than that needed to re-establish equilibrium. In the case of a quasi-static transformation, we can define the entropy at each instant because the system is infinitesimally close to equilibrium. This entropy, $S_{\text{tot}}(t)$, will be *a priori* time dependent and, for an isolated system, will be an increasing function of time.

An essential consequence of the principles mentioned above is that the entropy is a concave function of its arguments. Recall that a function, $f(x)$, is concave if for any x_1 and x_2 (Figure 1.4) we have

$$f\left(\frac{x_1 + x_2}{2}\right) \geq \frac{f(x_1) + f(x_2)}{2} \quad (1.8)$$

If a concave function is twice differentiable, its second derivative is either negative or zero: $f''(x) \leq 0$. For a convex function, $f''(x) \geq 0$.

Concavity of the entropy: *For a homogeneous system, the entropy is a concave function of the extensive variables (E, V, N).*

In effect, suppose that S is locally a strictly convex function of, for example, the energy in an interval around the value E ; we would then have

$$2S(E) < S(E - \Delta E) + S(E + \Delta E)$$

From additivity of the entropy, $2S(E) = S(2E)$: application of a constraint on the energy that renders the system inhomogeneous would allow entropy to increase. The maximum entropy principle would then lead to an inhomogeneous equilibrium

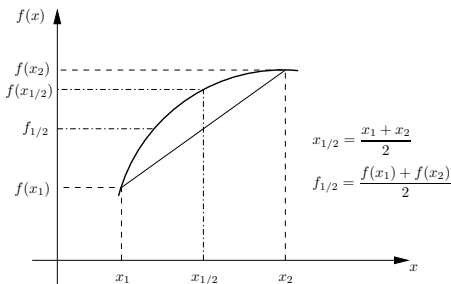


Figure 1.4 Properties of a concave function.

state, in contradiction with our starting hypothesis. Such behaviour is the signature of a phase transition (see Section 3.5.2). In our reasoning we have used a strict inequality, the case where the inequality is not strict will be examined soon. The generalization to several variables will be given in 1.4.1.

1.2.3 Intensive variables: temperature, pressure, chemical potential

We will now define the temperature, T , the pressure, P , and the chemical potential, μ , starting with the entropy, $S(E, V, N^{(1)}, \dots, N^{(r)})$. We shall later show that these definitions correspond to intuitive notions.

Temperature T

$$\left. \frac{\partial S}{\partial E} \right|_{V, N^{(i)}} = \frac{1}{T} \quad (1.9)$$

Pressure P

$$\left. \frac{\partial S}{\partial V} \right|_{E, N^{(i)}} = \frac{P}{T} \quad (1.10)$$

Chemical potential $\mu^{(i)}$

$$\left. \frac{\partial S}{\partial N^{(i)}} \right|_{E, V, N^{(j \neq i)}} = -\frac{\mu^{(i)}}{T} \quad (1.11)$$

The variables T , P and μ^i are called *intensive* variables: for example, if we multiply by two all the extensive variables, the temperature, pressure and chemical potential remain unchanged, which is clear from their definitions. On the other hand, the extensive variables, E , V and $N^{(i)}$ are doubled if we double the size of the system while keeping T , P and μ^i constant.

Temperature: thermal equilibrium

Let us first show that our definition of temperature allows us to arrive at the notion of thermal equilibrium. We use, again, the system depicted in Figure 1.3, where the piston is initially fixed, adiabatic and impermeable to molecules, and the two compartments (subsystems) are separately at equilibrium. In order to lighten the notations, and while assuming the wall to be impermeable to molecules, it will be convenient to define the entropy functions, $S_1(E_1, V_1)$ and $S_2(E_2, V_2)$, of the two subsystems by

$$S_1(E_1, V_1) = S(E_1, V_1, N_1^{(1)}, \dots, N_1^{(r)})$$

$$S_2(E_2, V_2) = S(E_2, V_2, N_2^{(1)}, \dots, N_2^{(r)})$$

Now make the piston diathermic: a new equilibrium will be established corresponding to a maximum of the entropy. All infinitesimal variations around this new equilibrium must obey the extremum condition $dS = 0$, where S is the total entropy

$$dS = \left. \frac{\partial S_1}{\partial E_1} \right|_{V_1} dE_1 + \left. \frac{\partial S_2}{\partial E_2} \right|_{V_2} dE_2 = 0 \quad (1.12)$$

The energy conservation condition, $E_1 + E_2 = E = \text{const}$, implies $dE_1 = -dE_2$, and, keeping in mind the definition (1.9), we conclude that the final temperatures, T_1 and T_2 , are equal, $T_1 = T_2 = T$. When energy is allowed to flow freely in the form of heat, the final equilibrium corresponds to the temperatures of the two subsystems being equal, that is to say thermal equilibrium.

We now show that energy flows from the hot to the cold compartment. Figure 1.5 shows $\partial S/\partial E$ for the two subsystems (i.e. $1/T_1$ and $1/T_2$) as a function of E . Because $S(E)$ is concave, the two curves, $\partial S_1/\partial E_1$ and $\partial S_2/\partial E_2$ are decreasing functions of E_1 and E_2 respectively. We have assumed for the initial temperatures that $1/T'_1 > 1/T'_2$, i.e. $T'_1 < T'_2$. In other words, compartment (1) is initially colder than compartment (2). Suppose that the final energy, E_1 , of (1) is smaller than its initial energy ($E_1 < E'_1$), and thus $T_1 < T'_1$. From energy conservation, we have $E_2 > E'_2$ and, therefore, $T_2 > T'_2$. It is then impossible to have equal temperatures and the final state does not satisfy the maximum entropy principle. Equality of temperatures is possible only if $E_1 > E'_1$ and $E_2 < E'_2$, that is only when energy flows from the hotter to the colder compartment. This property conforms to our intuitive idea of temperature and heat.

We have so far assumed that $\partial^2 S/\partial E^2$ is strictly negative, i.e. that $\partial S/\partial E$ is strictly decreasing. It can happen that $\partial^2 S/\partial E^2 = 0$ over an interval in E , in which case $S(E)$ has a linear part (Figure 1.6). This means that an energy transfer does

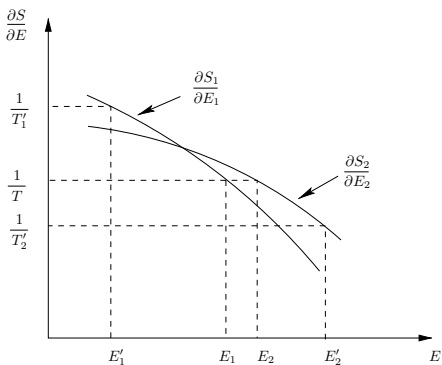


Figure 1.5 Graphic illustration of energy flow.

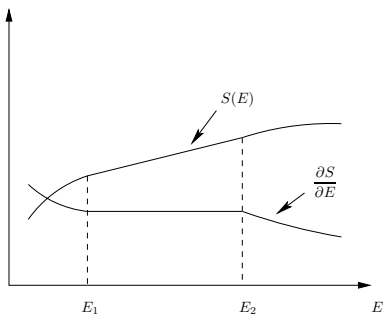


Figure 1.6 The entropy and its derivative in the presence of a phase transition.

not change the temperature. Such a situation arises at a phase transition: when we supply heat to a mixture of ice and water, we first melt the ice without changing its temperature. In such a case, the system is no longer homogeneous: in order to characterize it, one has to give, in addition to the temperature, the fraction of the system in each of the phases.

Pressure: mechanical equilibrium

Once again, we start with the situation shown in Figure 1.3, but now we make the piston both diathermic and mobile. Once equilibrium is reached, the entropy variation, dS , should vanish for all small fluctuations of the energy and volume

$$dS = \left(\frac{\partial S_1}{\partial E_1} \Big|_{V_1} dE_1 + \frac{\partial S_2}{\partial E_2} \Big|_{V_2} dE_2 \right) + \left(\frac{\partial S_1}{\partial V_1} \Big|_{E_1} dV_1 + \frac{\partial S_2}{\partial V_2} \Big|_{E_2} dV_2 \right) = 0$$

The conservation conditions of energy, $E_1 + E_2 = E = \text{const}$, and volume, $V_1 + V_2 = V = \text{const}$, along with the definitions (1.9) and (1.10) imply

$$dS = \left(\frac{1}{T_1} - \frac{1}{T_2} \right) dE_1 + \left(\frac{P_1}{T_1} - \frac{P_2}{T_2} \right) dV_1 = 0 \quad (1.13)$$

Since the energy and volume fluctuations are independent to first order in (dE, dV) (see Footnote 13), the extremum condition for the entropy, $dS = 0$, implies thermal equilibrium, $T_1 = T_2 = T$, and mechanical equilibrium, $P_1 = P_2 = P$. The connection between $(\partial S / \partial V)_E$ and the usual concept of pressure will be made very soon.

The reader will remark that the case where the piston is mobile and adiabatic is not symmetric with that when the piston is diathermic and fixed: a variation in the volume causes a transfer of energy, whereas a fixed piston prevents the exchange of volume. If the piston is made mobile but not diathermic, it will oscillate indefinitely in the absence of friction. If we take into account the friction of the piston against the cylinder and the viscosities of the two fluids, we obtain a final stationary situation where the temperatures of the two compartments are *a priori* different and are determined by the viscosities and friction.

Returning to Equation (1.13) and assuming that we are in the neighbourhood of an equilibrium with $T_1 = T_2 = T$ and P_1 slightly larger than P_2 , the variation, dV_1 , has to be such that the entropy increases ($dS \geq 0$), which implies that $dV_1 > 0$. In other words, the volume of the compartment that was at higher initial pressure increases.¹³

¹³ The alert reader will remark that the final equilibrium temperature will be slightly different from T . However, the change in temperature, calculated explicitly for a van der Waals gas in Problem 1.7.2, is of the order of $(dV_1)^2$ and so is dE_1 . On the other hand, for finite variations, energy and volume are not independent parameters.

Chemical potential: equilibrium of particle (or molecule) flux

Finally, the piston is made diathermic, fixed, and permeable to molecules of type (i) . Returning to the general notation $S(E_m, V_m, N_m^{(1)}, \dots, N_m^{(r)})$, we again use the principle of maximum entropy

$$\begin{aligned} dS = & \left(\frac{\partial S}{\partial E_1} \Big|_{V_1, N_1^{(i)}} dE_1 + \frac{\partial S}{\partial E_2} \Big|_{V_2, N_2^{(i)}} dE_2 \right) \\ & + \left(\frac{\partial S}{\partial N_1^{(i)}} \Big|_{E_1, V_1, N_1^{(j \neq i)}} dN_1^{(i)} + \frac{\partial S}{\partial N_2^{(i)}} \Big|_{E_2, V_2, N_2^{(j \neq i)}} dN_2^{(i)} \right) = 0 \end{aligned}$$

The conservation laws for the energy and the number of molecules of type (i) give $dE_1 = -dE_2$ and $dN_1^{(i)} = -dN_2^{(i)}$, and using the definitions (1.9) and (1.11) we get

$$dS = \left(\frac{1}{T_1} - \frac{1}{T_2} \right) dE_1 - \left(\frac{\mu_1^{(i)}}{T_1} - \frac{\mu_2^{(i)}}{T_2} \right) dN_1^{(i)} = 0 \quad (1.14)$$

The entropy being stationary implies that at equilibrium the chemical potentials are equal: the transfer of molecules between the two subsystems stops at equilibrium. The term ‘chemical potential’ is used for historical reasons, but this terminology is unfortunate because we see clearly in our example that no chemical reaction is involved in its definition. It is easy to generalize this to all types of molecules present. The condition for equilibrium is¹⁴

$$\mu_1^{(1)} = \mu_2^{(1)}, \dots, \mu_1^{(i)} = \mu_2^{(i)}, \dots, \mu_1^{(r)} = \mu_2^{(r)}$$

Reasoning along the same lines as for the pressure allows us to show that the number of particles decreases in the compartment where the chemical potential was the larger.

Equation of state

Assuming $N^{(i)}$ to be constant, the definitions (1.9) and (1.10) of T and P define the functions f_T and f_P of E and V

$$\frac{1}{T} = \frac{\partial S}{\partial E} \Big|_V = f_T(E, V) \quad \frac{P}{T} = \frac{\partial S}{\partial V} \Big|_E = f_P(E, V)$$

Since f_T is a strictly decreasing function of E for a homogeneous system, we can determine E as a function of T at fixed V , $E = g(T, V)$, and substituting in the

¹⁴ We assume here that no chemical reactions take place when the subsystems are put in contact. This case will be studied in Chapter 3.

equation for the pressure gives

$$P = T f_P(g(T, V), V) = h(T, V) \quad (1.15)$$

The relation $P = h(T, V)$ is the *equation of state* for a homogeneous substance. The best known equation of state is that of the *ideal gas* written here in three useful forms

$$\boxed{P = \frac{RT}{V} = \frac{\mathcal{N}kT}{V} = nkT} \quad (1.16)$$

This equation is written for one mole of gas, R is the ideal gas constant, $\mathcal{N}$ is Avogadro's number, $k = R/\mathcal{N}$ is the Boltzmann constant and $n = \mathcal{N}/V$ is the density of molecules.

1.2.4 Quasi-static and reversible processes

Putting together the definitions (1.9) to (1.11) and considering only one kind of molecules, we can write

$$dS = \frac{1}{T} dE + \frac{P}{T} dV - \frac{\mu}{T} dN \quad (1.17)$$

or

$$\boxed{T dS = dE + P dV - \mu dN} \quad (1.18)$$

which is the ' $T dS$ equation in the entropy variable', referring to the fact that the basic function is $S(E, V, N)$. But since S is an increasing function of E , the correspondence between them is one-to-one, which allows us to find an energy function $E(S, V, N)$. The $T dS$ equation can now be written in the equivalent form

$$\boxed{dE = T dS - P dV + \mu dN} \quad (1.19)$$

This is the ' $T dS$ equation in the energy variable'. When $dV = 0$ (fixed piston) and $dN = 0$ (no molecule transfer), the energy transferred is in the form of heat. We can then make the identification: $dE = T dS = \delta Q$.

At this point, it is important to make a crucial remark: we have emphasized that the existence of the entropy function supposes a situation of equilibrium, or infinitesimally close to it, and that the $T dS$ equations are *a priori* valid only for quasi-static processes. In particular, the equation $\delta Q = T dS$ is also valid only for a quasi-static process. Similarly, let us re-consider the expression for mechanical work given by Equation (1.3): if the area of the piston is A , the force is given by $F = P_{\text{ext}}A$ where P_{ext} is the external pressure applied by the gas (if the force is due to an external pressure). In a quasi-static process, the external pressure, P_{ext} ,

is infinitesimally close to the internal pressure, P . Consequently,

$$\delta W = F dx = P_{\text{ext}} A dx = -P dV \quad (1.20)$$

The minus sign is due to the fact that we supply work by compressing the gas: $dx > 0$ corresponds to $dV < 0$ (Figure 1.1(a)). Note that the first two equalities giving δW are valid in general, but that the last is valid only for a quasi-static process since the pressure, P , of a gas is only defined at equilibrium (see Problem 1.7.1). By comparing with (1.5) we see that the quantity $(-\partial E/\partial V)_{S,N}$ is indeed the pressure, as is $T(\partial S/\partial V)_{E,N}$.

We emphasize, yet again, the essential point that (for N fixed) conservation of energy or the ‘first law’, $dE = \delta Q + \delta W$, is always valid. The equation $dE = T dS + \delta W$, which relies on the notion of entropy and, therefore, the ‘second law’, is valid only for quasi-static processes. For such processes, δW can be expressed as a function of variables that are internal to the system with no reference to the characteristics of the external medium. A good starting point to solve a thermodynamic problem is to write the $T dS$ equation adapted to the specific problem under consideration.

We finally introduce the notion of *reversible transformation*: by definition, a quasi-static transformation is reversible if it takes place at constant total entropy. In general, lifting one or more internal constraints in an isolated system increases the entropy, $S_{\text{final}} > S_{\text{initial}}$. Re-imposing these constraints cannot return the system to its initial state. The transformation is, therefore, *irreversible*. For a strictly isolated system, evolution takes place spontaneously in a certain direction, with the entropy increasing until equilibrium is reached. On the other hand, for a reversible transformation, we can return the system to its initial state by choosing appropriately the manner in which we change the internal constraints. This explains the term ‘reversible transformation’: it can be effected in both directions. By means of external intervention, a small modification of the constraints fixes the sense of the reversible transformation. For example, we can choose the sense of the displacement of the piston by increasing slightly the pressure (or adding energy¹⁵) in one of the compartments of Figure 1.3. We note that a transformation that is, at the same time, quasi-static and takes place without any energy exchange in the form of heat, is reversible. In fact, since the transformation is quasi-static, all energy exchange in the form of heat can be written as $\delta Q = T dS$, and since $\delta Q = 0$, this means that $dS = 0$. We have already seen that all transformations of an isolated system produce an entropy change $\Delta S \geq 0$. On the other hand, it can happen that parts of an isolated system experience a decrease in entropy, as long as this is compensated by an increase in S in other parts of the system.

¹⁵ We remark that, if a system at a temperature $T - dT$ and with specific heat C absorbs, in the form of heat, a quantity of energy $\delta Q = C dT$ from a reservoir at T , the entropy change is proportional to $(dT)^2$.

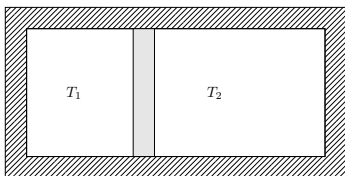


Figure 1.7 Example of a quasi-static but non-reversible process: two compartments separated by an almost adiabatic partition.

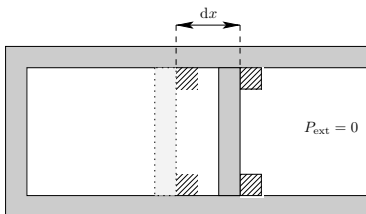


Figure 1.8 Example of an infinitesimal but not quasi-static process.

It is useful to give an example of a quasi-static transformation that is *not* reversible. Consider Figure 1.7: the two compartments are at different temperatures, but the wall is almost adiabatic (heat flow is very slow) and each compartment is infinitesimally close to equilibrium. The entropy, S , is well defined but increases with time (Section 2.6). As we shall see in Chapter 6, the wall is in a state of local equilibrium because the temperature is position dependent but well defined at all points and entropy production takes place in this wall.

It is also useful to keep in mind that an infinitesimal transformation is not necessarily quasi-static. In the example of Figure 1.8, the container and the piston are assumed to be adiabatic: the piston is first released and moves an infinitesimal distance, dx , before being blocked. Assuming the gas is ideal, the entropy change is $dS = R dV/V$ for one mole of gas (see Problem 1.7.1). The transformation is adiabatic, and were it quasi-static it would also be reversible, which is not the case since $dS \geq 0$. The sudden movement of the piston creates turbulence in its neighbourhood with the consequence that the transformation no longer is near

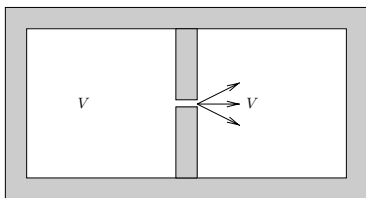


Figure 1.9 An infinitely slow process which is not quasi-static.

equilibrium. Entropy creation comes from viscous friction, which returns the gas to equilibrium. Similarly, in the example of Figure 1.9, where the container is adiabatic, the compartment on the right is initially empty and gas (presumed ideal) leaks into it through an infinitesimal hole: at equilibrium, the two compartments are equally full and the entropy change is $\Delta S = R \ln 2$ for two identical compartments and one mole of gas. Although the transformation is infinitely slow, it does not take place through a succession of equilibrium states. Problem 1.7.1 gives a detailed comparison between reversible and irreversible adiabatic transformations.

1.2.5 Maximum work and heat engines

In the usual presentation of thermodynamics, a central rôle is played by the heat engine. A device, $\mathcal{X}$, connected to two heat sources¹⁶ at temperatures T_1 and T_2 , $T_1 > T_2$, supplies work to the outside (Figure 1.10(a)). We establish first a preliminary result: *the theorem of maximum work*. Suppose that $\mathcal{X}$ receives a quantity of energy,¹⁷ Q , from a heat source, S , at temperature T . We assume that S is big enough for Q to be infinitesimal, compared to its total energy, so that its temperature, T , stays essentially unchanged (Figure 1.10(b)). The transformation experienced by the source is quasi-static and its entropy change is $-Q/T$: such a source is called a *heat reservoir*. The system, $\mathcal{X}$, supplies an amount of work, $\mathbb{W}$, to the outside world, symbolized by a spring in Figure 1.10(b). This spring is purely a mechanical object that does not contribute to the entropy. The combination $[\mathcal{X} + S]$ is thermally isolated, but the wall separating $\mathcal{X}$ from S is rendered diathermic for a time period during which $\mathcal{X}$ and S come in thermal contact. This thermal contact and the work done will cause $\mathcal{X}$ to go from one equilibrium state to

¹⁶ By heat source we mean a source that can exchange energy only in the form of heat.

¹⁷ We should be careful with the signs: the source receives a quantity of heat $-Q$, the system receives an amount of work $W = -\mathbb{W}$.

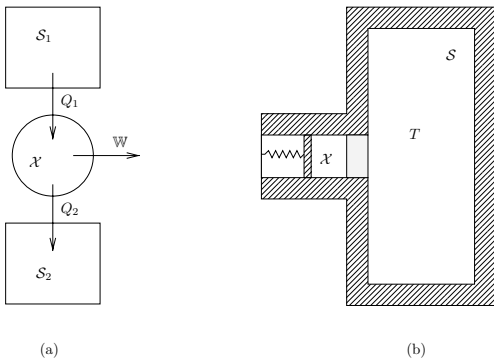


Figure 1.10 (a) Heat engine, (b) illustration of the theorem of maximum work.

another, and with this transformation its energy will change by ΔE and its entropy by ΔS . Conservation of energy gives the following expression for the *supplied* work $\mathbb{W}$

$$\mathbb{W} = -\Delta E + Q$$

The work, $\mathbb{W}$, is maximum if Q is maximum. But, the maximum entropy principle applied to the isolated system $[\mathcal{X} + S]$ implies

$$\Delta S_{\text{tot}} = \Delta S - \frac{Q}{T} \geq 0$$

From this the theorem of maximum work follows

$$\boxed{\mathbb{W} \leq T \Delta S - \Delta E} \quad (1.21)$$

Maximum work is obtained from a transformation when it is reversible and $\Delta S_{\text{tot}} = 0$.

The device $\mathcal{X}$ functions in a cycle, that is to say, it returns periodically to the same state. Let $\mathbb{W}$ be the work supplied¹⁸ by $\mathcal{X}$ during one such cycle, Q_1 the

¹⁸ We will often use the term 'work (or heat) supplied' as shorthand for energy supplied in the form of work (or heat).

heat given to $\mathcal{X}$ by the hot reservoir and Q_2 the heat given by $\mathcal{X}$ to the cold reservoir. During this cyclical process, $\mathcal{X}$ is in contact successively with the cold and hot reservoirs. The transformations taking place during a cycle are assumed to be reversible, in order to obtain the maximum work. The entropy changes of the two sources are $-Q_1/T_1$ and Q_2/T_2 respectively. The entropy of $\mathcal{X}$ does not change since it returns to its initial state at the end of a cycle. We therefore have

$$-\frac{Q_1}{T_1} + \frac{Q_2}{T_2} = 0$$

Due to energy conservation, the work done during a cycle will be equal to $Q_1 - Q_2$, and

$$\mathbb{W} = Q_1 - Q_2 = Q_1 \left(1 - \frac{T_2}{T_1} \right) \quad (1.22)$$

We deduce from this the second law in its usual form: if $T_1 = T_2$, no work is done and we cannot obtain work by a cyclical transformation using only one heat source. The thermal efficiency, η , of a heat engine is defined as the fraction of Q_1 that is changed into work. For the given temperatures, T_1 and T_2 , this efficiency is maximum for an engine functioning reversibly and gives, from Equation (1.22),

$$\boxed{\eta = 1 - \frac{T_2}{T_1}} \quad (1.23)$$

Entropy is defined up to a multiplicative constant: replacing S by λS , where λ is the same constant for all systems, preserves the fundamental extremum properties. With this transformation, $T \rightarrow T/\lambda$ as given by the definition (1.9). We fix λ by demanding that the temperature, in kelvins (K), of the triple point of water be equal to 273.16. Since energy is measured in joules (J), entropy is measured in J K^{-1} . The temperature, $\mathcal{T}$, measured on the Celsius scale is defined by $\mathcal{T} = T - 273.15 \text{ K}$.

1.3 Thermodynamic potentials

1.3.1 Thermodynamic potentials and Massieu functions

Assuming for the moment N to be constant, the energy, E , is then a function of S and V . In thermodynamics, it is very often convenient to be able to change variables and use, for example, the temperature and pressure rather than the entropy and volume. These variable changes are implemented with the help of thermodynamic potentials or Massieu functions that are, respectively, the Legendre transforms (Section A.1) of the energy and entropy. We start with the energy and show how to go from a function of entropy to one of temperature. From Equation (1.19)

we have

$$\left. \frac{\partial E}{\partial S} \right|_V = T$$

The Legendre transform, F , of E with respect to S is given by

$$F(T, V) = E - TS \quad \left. \frac{\partial F}{\partial T} \right|_V = -S, \quad \left. \frac{\partial F}{\partial V} \right|_T = -P \quad (1.24)$$

The function F is called the *free energy*, and going from E to F has allowed us to use the temperature instead of the entropy. The two other thermodynamic potentials are the *enthalpy* $\bar{H}$, which is the Legendre transform of the energy with respect to the volume,

$$\bar{H}(S, P) = E + PV \quad \left. \frac{\partial \bar{H}}{\partial S} \right|_P = T, \quad \left. \frac{\partial \bar{H}}{\partial P} \right|_S = V \quad (1.25)$$

and the *Gibbs potential*, which is obtained by a double Legendre transform

$$G(T, P) = E - TS + PV \quad \left. \frac{\partial G}{\partial T} \right|_P = -S, \quad \left. \frac{\partial G}{\partial P} \right|_T = V \quad (1.26)$$

The *Maxwell relations* are an important consequence of the existence of these functions. In fact, the equality

$$\frac{\partial^2 F}{\partial T \partial V} = \frac{\partial^2 F}{\partial V \partial T}$$

combined with Equation (1.24), implies

$$\left. \frac{\partial S}{\partial V} \right|_T = \left. \frac{\partial P}{\partial T} \right|_V \quad (1.27)$$

whereas the same reasoning applied to the Gibbs potential gives

$$\left. \frac{\partial S}{\partial P} \right|_T = - \left. \frac{\partial V}{\partial T} \right|_P \quad (1.28)$$

Instead of taking the Legendre transforms of the energy, we can take those of the entropy and thus obtain the Massieu functions. For example, from Equation (1.17) we have

$$\left. \frac{\partial S}{\partial E} \right|_V = \frac{1}{T}$$

and the Massieu function Φ_1 is obtained by taking the Legendre transform with respect to the energy

$$\begin{aligned}\Phi_1\left(\frac{1}{T}, V\right) &= S - \frac{E}{T} = -\frac{1}{T} F \\ \left.\frac{\partial \Phi_1}{\partial 1/T}\right|_V &= -E \quad \left.\frac{\partial \Phi_1}{\partial V}\right|_{1/T} = \frac{P}{T}\end{aligned}\tag{1.29}$$

The reader can easily construct the two other Massieu functions (Exercise (1.6.1)).

For historical reasons in thermodynamics, the variable T is more commonly used than the variable $1/T$ and the energy representation (1.19) is preferred to that of the entropy (1.17), and F to Φ_1 . However, since entropy is of a different nature from the mechanical variables E and V , the entropy representation would be the more natural. We shall also see that the natural variable in statistical mechanics is $1/T$, that Φ_1 is the logarithm of the partition function, and that the entropy representation imposes itself in out-of-equilibrium situations.

1.3.2 Specific heats

Suppose we give a system a quantity of heat, δQ , in a quasi-static process, while keeping fixed one or more thermodynamic variables, y . If dT is the increase in temperature, the *specific heat* (or heat capacity), C_y , at fixed y , is the ratio

$$C_y = \left.\frac{\delta Q}{dT}\right|_y = T \left.\frac{\partial S}{\partial T}\right|_y\tag{1.30}$$

The substitution of $T dS$ for δQ is justified because we have assumed a quasi-static process. The classic cases are the specific heat at constant volume, C_V ,

$$\boxed{C_V = \left.\frac{\delta Q}{dT}\right|_V = T \left.\frac{\partial S}{\partial T}\right|_V = \left.\frac{\partial E}{\partial T}\right|_V}\tag{1.31}$$

and the specific heat at constant pressure, C_P ,

$$\boxed{C_P = \left.\frac{\delta Q}{dT}\right|_P = T \left.\frac{\partial S}{\partial T}\right|_P = \left.\frac{\partial \overline{H}}{\partial T}\right|_P}\tag{1.32}$$

It is the enthalpy, $\overline{H}$, and not the energy, that appears in this last equation!

We now present a classic calculation leading to a useful relation between the specific heats at constant volume and pressure. It is helpful to define the following three coefficients which we define as intensive variables.

Expansion coefficient at constant pressure:

$$\alpha = \frac{1}{V} \left. \frac{\partial V}{\partial T} \right|_P \quad (1.33)$$

Coefficient of isothermal compressibility:

$$\kappa_T = -\frac{1}{V} \left. \frac{\partial V}{\partial P} \right|_T \quad (1.34)$$

Coefficient of adiabatic compressibility (or at constant entropy):

$$\kappa_S = -\frac{1}{V} \left. \frac{\partial V}{\partial P} \right|_S \quad (1.35)$$

We need the following relation between partial derivatives. Consider a function of two variables, $z(x, y)$, and its differential

$$dz = \left. \frac{\partial z}{\partial x} \right|_y dx + \left. \frac{\partial z}{\partial y} \right|_x dy$$

If we now restrict ourselves to a surface $z = \text{const}$, i.e. $dz = 0$, we get

$$\left. \frac{\partial z}{\partial x} \right|_y dx = - \left. \frac{\partial z}{\partial y} \right|_x dy$$

or, in a form that is easy to write by circular permutation of the three variables (x, y, z) ,

$$\boxed{\left. \frac{\partial x}{\partial y} \right|_z \left. \frac{\partial y}{\partial z} \right|_x \left. \frac{\partial z}{\partial x} \right|_y = -1} \quad (1.36)$$

Applying this relation to the variables (T, P, V)

$$\left. \frac{\partial T}{\partial P} \right|_V \left. \frac{\partial P}{\partial V} \right|_T \left. \frac{\partial V}{\partial T} \right|_P = -1 \quad (1.37)$$

We start with $T dS$ expressed in terms of the variables (T, P)

$$T dS = C_P dT + T \left. \frac{\partial S}{\partial P} \right|_T dP = C_P dT - T \left. \frac{\partial V}{\partial T} \right|_P dP = C_P dT - TV\alpha dP$$

where we have first used the Maxwell relation (1.28) and then the definition of α (1.33). From this we find, by going from variables (T, P) to (T, V) ,

$$T \left. \frac{\partial S}{\partial T} \right|_V = C_P - TV\alpha \left. \frac{\partial P}{\partial T} \right|_V = C_P - TV\alpha \frac{\alpha}{\kappa_T}$$

The second equality has been obtained by evaluating $(\partial P/\partial T)_V$ with the help of (1.37). The final result is the classic relation for $C_P - C_V$

$$C_P - C_V = \frac{TV\alpha^2}{\kappa_T} \quad (1.38)$$

Another relation is (Exercise 1.6.3)

$$\frac{C_P}{C_V} = \frac{\kappa_T}{\kappa_S} \quad (1.39)$$

1.3.3 Gibbs–Duhem relation

We now consider a situation where the number of particles, N , can change. If we scale all the extensive variables by a factor λ

$$E \rightarrow \lambda E \quad V \rightarrow \lambda V \quad N \rightarrow \lambda N$$

the entropy, being also extensive, will scale as $S \rightarrow \lambda S$, and consequently

$$\lambda S = S(\lambda E, \lambda V, \lambda N)$$

By differentiating with respect to λ , using (1.17), and then taking $\lambda = 1$, we obtain

$$S = \frac{E}{T} + \frac{PV}{T} - \frac{\mu N}{T}$$

which allows us to identify μN as the Gibbs potential, G (1.26)

$$\mu N = E - TS + PV = G \quad (1.40)$$

The differential of the Gibbs potential can be written in two ways by using (1.26) or (1.40)

$$dG = -S dT + V dP + \mu dN = \mu dN + N d\mu$$

from which is obtained the ‘Gibbs–Duhem’ relation

$$N d\mu + S dT - V dP = 0 \quad (1.41)$$

We apply this to the isothermal case, $dT = 0$. With the particle density defined by $n = N/V$, the Gibbs–Duhem relation yields

$$\left. \frac{\partial P}{\partial \mu} \right|_T = n$$

We now calculate $(\partial\mu/\partial n)_T$ with the help of definition (1.34) for the isothermal compressibility, κ_T , and with $v = V/N = 1/n$

$$\frac{1}{\kappa_T} = -v \left. \frac{\partial P}{\partial v} \right|_T = n \left. \frac{\partial P}{\partial n} \right|_T = n \left. \frac{\partial P}{\partial \mu} \right|_T \left. \frac{\partial \mu}{\partial n} \right|_T = n^2 \left. \frac{\partial \mu}{\partial n} \right|_T$$

and therefore

$$\left. \frac{\partial \mu}{\partial n} \right|_T = \frac{1}{n^2 \kappa_T} \quad (1.42)$$

1.4 Stability conditions

1.4.1 Concavity of entropy and convexity of energy

We now generalize to several variables the concavity analysis of the entropy that we did in 1.2.2 for one variable. To simplify the notation, we limit ourselves to two variables, the energy E and volume V , but the generalization to more variables is straightforward. Let $S(2E, 2V)$ be the entropy of a homogeneous, isolated system at equilibrium with energy $2E$ and volume $2V$. We now divide the system into two subsystems with energies $(E \pm \Delta E)$ and volumes $(V \pm \Delta V)$, and we suppose that the entropy is locally convex,

$$S(E + \Delta E, V + \Delta V) + S(E - \Delta E, V - \Delta V) > 2S(E, V) = S(2E, 2V)$$

In this case, applying an internal constraint would render the system inhomogeneous (unless $\Delta E/E = \Delta V/V$) while allowing the entropy to increase. The principle of maximum entropy would then lead to an inhomogeneous equilibrium state in contradiction with the initial assumption. We obtain the condition that the entropy must be concave

$$S(E + \Delta E, V + \Delta V) + S(E - \Delta E, V - \Delta V) \leq 2S(E, V) \quad (1.43)$$

This inequality can also be written in the form

$$\boxed{(\Delta S)_{(E,V)} \leq 0} \quad (1.44)$$

which can be interpreted as follows. At fixed E and V , internal constraints can only decrease the entropy. Taking the Taylor expansion of $S(E \pm \Delta E, V \pm \Delta V)$ for small ΔE and ΔV , we get

$$\begin{aligned} S(E \pm \Delta E, V \pm \Delta V) &\simeq S(E, V) \pm \Delta E \frac{\partial S}{\partial E} \pm \Delta V \frac{\partial S}{\partial V} + \frac{1}{2} (\Delta E)^2 \frac{\partial^2 S}{\partial E^2} \\ &\quad + \frac{1}{2} (\Delta V)^2 \frac{\partial^2 S}{\partial V^2} + \Delta E \Delta V \frac{\partial^2 S}{\partial E \partial V} \end{aligned}$$

Substituting this expansion in Equation (1.43) we obtain the inequality

$$(\Delta E)^2 \frac{\partial^2 S}{\partial E^2} + (\Delta V)^2 \frac{\partial^2 S}{\partial V^2} + 2\Delta E \Delta V \frac{\partial^2 S}{\partial E \partial V} \leq 0 \quad (1.45)$$

The concavity condition on the entropy can be transformed into a convexity condition on the energy. Consider an isolated system with energy E and volume V , which we divide into two subsystems at equilibrium

$$E = E_1 + E_2 \quad V = V_1 + V_2$$

We now apply an internal constraint where

$$E_1 \rightarrow E_1 + \Delta E \quad E_2 \rightarrow E_2 - \Delta E$$

and

$$V_1 \rightarrow V_1 + \Delta V \quad V_2 \rightarrow V_2 - \Delta V$$

The principle of maximum entropy gives

$$S(E_1 + \Delta E, V_1 + \Delta V) + S(E_2 - \Delta E, V_2 - \Delta V) \leq S(E, V)$$

But S is an increasing function of the energy. There exists, therefore, an energy $\tilde{E} \leq E$ such that

$$S(E_1 + \Delta E, V_1 + \Delta V) + S(E_2 - \Delta E, V_2 - \Delta V) = S(\tilde{E}, V)$$

At constant entropy, internal constraints can only increase the energy: $E_1 + E_2 = E \geq \tilde{E}$. Therefore, the analogue of Equation (1.44) for the energy is

$$\boxed{(\Delta E)_{(S,V)} \geq 0} \quad (1.46)$$

which says that the energy is a convex function of S and V . Another demonstration of this result is proposed in Exercise 1.6.2.

1.4.2 Stability conditions and their consequences

The convexity condition on the energy leads to an equation analogous to (1.45). Only the sense of the inequality changes when we go from concavity of the entropy to convexity of the energy

$$(\Delta S)^2 \frac{\partial^2 E}{\partial S^2} + (\Delta V)^2 \frac{\partial^2 E}{\partial V^2} + 2\Delta S \Delta V \frac{\partial^2 E}{\partial S \partial V} \geq 0 \quad (1.47)$$

This condition can be conveniently expressed in matrix form by introducing the symmetric 2×2 matrix $\mathcal{E}$ whose elements are second derivatives of E

$$\mathcal{E} = \begin{pmatrix} \frac{\partial^2 E}{\partial S^2} & \frac{\partial^2 E}{\partial S \partial V} \\ \frac{\partial^2 E}{\partial V \partial S} & \frac{\partial^2 E}{\partial V^2} \end{pmatrix} = \begin{pmatrix} E''_{SS} & E''_{SV} \\ E''_{VS} & E''_{VV} \end{pmatrix} \quad (1.48)$$

Introducing the two-component vector, $x = (\Delta S, \Delta V)$, and its transpose, x^T , Equation (1.47) becomes $x^T \mathcal{E} x \geq 0$, which means that the matrix $\mathcal{E}$ must be *positive*. By definition, a symmetric¹⁹ (and therefore diagonalizable) $N \times N$ matrix, A_{ij} , is said to be positive if for any vector x with components x_i , $i = 1, \dots, N$

$$x^T A x = \sum_{i,j=1}^N x_i A_{ij} x_j \geq 0 \quad (1.49)$$

The positivity condition for a matrix can be expressed in terms of its eigenvalues. A necessary and sufficient condition for a matrix to be positive is that all its eigenvalues, λ_i , be positive, $\lambda_i \geq 0$. For a positive definite matrix, this inequality, and those in Equation (1.49), become strict inequalities. In fact, we can diagonalize a symmetric matrix by $A = R^T \Lambda R$, where Λ is a diagonal matrix and R an orthogonal matrix $R^T = R^{-1}$. The positivity condition (1.49) becomes

$$y^T \Lambda y = \sum_{i=1}^N \lambda_i y_i^2 \geq 0$$

where the N -component vector y is given by $y = Rx$. Clearly, this implies that $\lambda_i \geq 0$. We return now to the case of a 2×2 matrix,

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix} \quad (1.50)$$

An elementary calculation gives the positivity condition. For the two eigenvalues to be positive, we must have $a + c \geq 0$ and $ac - b^2 \geq 0$, which means that $a \geq 0$ and $c \geq 0$ separately. In the case of a negative matrix we have $a \leq 0$ and $c \leq 0$ whereas $ac - b^2 \geq 0$.

The analysis of the stability conditions is most conveniently done with the help of the thermodynamic potentials. The Legendre transform changes concavity to convexity (Section A.1), and consequently we have

$\overline{H}$	convex in E	concave in P
F	concave in T	convex in V
G	concave in T	concave in P

¹⁹ For real matrices one needs to impose the symmetry condition, but in a complex vector space, a positive matrix is automatically Hermitian and thus diagonalizable.

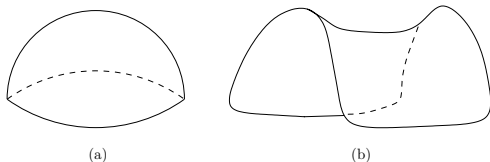


Figure 1.11 Surfaces with (a) positive and (b) negative curvatures.

Geometrically, the curvature K of a surface, $z = f(x, y)$, is given by

$$K = \frac{f''_{xx}f''_{yy} - (f''_{xy})^2}{(1 + f'^2_x + f'^2_y)^2} \quad (1.51)$$

E , S , and G are therefore positively curved functions of their respective variables (sphere-like shape) whereas $\bar{H}$ and F are negatively curved functions (saddle-like shape), see Figure 1.11.

The stability conditions are obtained most simply by studying the Gibbs potential. The matrix constructed from the second derivatives, G''_{TT} , G''_{TP} , G''_{PP} must be negative since G is a concave function²⁰ of T and P . We therefore have

$$G''_{TT} = - \left. \frac{\partial S}{\partial T} \right|_P = - \frac{C_P}{T} \leq 0 \Rightarrow C_P \geq 0$$

and

$$G''_{PP} = \left. \frac{\partial V}{\partial P} \right|_T = -V\kappa_T \leq 0 \Rightarrow \kappa_T \geq 0$$

where κ_T is the isothermal compressibility defined in (1.34). The last condition is

$$G''_{TT}G''_{PP} - (G''_{TP})^2 \geq 0$$

Using $G''_{PT} = (\partial V / \partial T)_P = \alpha V$, where α is the constant pressure expansion coefficient defined in (1.33), the above condition can be written as

$$C_P - \frac{\alpha^2 V T}{\kappa_T} \geq 0$$

Equation (1.38) leads to the conclusion that $C_V \geq 0$. To summarize, the two stability conditions are

$$\boxed{C_V \geq 0 \quad \kappa_T \geq 0} \quad (1.52)$$

²⁰ In Section A.1.2 we show that G is indeed a concave function of both variables T and P .

With Equations (1.38) and (1.39) in mind, the stability conditions (1.52) yield $C_P \geq C_V \geq 0$ and $\kappa_T \geq \kappa_S \geq 0$. The condition $C_V \geq 0$ can be obtained directly from the concavity of the entropy, but the condition on κ_T is more difficult to show directly from S . Although one might think intuitively that the expansion coefficient, α , must be positive (an increase in temperature normally leads to expansion, not contraction), we should note that the stability conditions do not impose any restrictions on this coefficient. In fact, it can happen that α becomes negative!

1.5 Third law of thermodynamics

1.5.1 Statement of the third law

The ‘third law’ is fundamentally related to low temperatures. Before stating it, we give orders of magnitude of temperatures reached by various techniques²¹

Pumped helium-4	1 K
Mixture helium-3–helium-4	10 mK
Helium-3 compression (Problem 5.7.6)	2 mK
Electronic spin demagnetization (Problem 1.7.7)	3 mK
Nuclear spin demagnetization	10 μ K
Bose–Einstein condensates of atomic gases (Problem 5.7.5)	1 nK

Laser cooling also produces nK temperatures, but there is no thermal equilibrium here, only an effective temperature. The temperature of an atomic Bose–Einstein condensate corresponds to metastable equilibrium, moreover these condensates cannot be used to cool other systems.

The third law of thermodynamics allows us to fix the entropy at zero temperature: ‘The entropy tends to zero as the temperature vanishes.’ More precisely, for a system of volume V and entropy $S(V)$ ²²

$$\lim_{T \rightarrow 0} \lim_{V \rightarrow \infty} \frac{1}{V} S(V) = 0 \quad (1.53)$$

The first limit in Equation (1.53) corresponds to the *thermodynamic limit*, the intensive quantities remain finite.

Current techniques allow cooling down to the mK scale. At that temperature there remains the residual entropy due to the nuclear spins, which vanishes only

²¹ The method of cooling by nuclear spin demagnetization can reach temperatures of 0.3 nK, but this is the temperature of the spin lattice, which is very different in nature from the usual temperature, it can even be negative. See Problem 3.8.2.

²² There are, however, model systems which retain a non-vanishing entropy at zero temperature. It has been known for a long time that the antiferromagnetic Ising model on a two-dimensional triangular lattice has an entropy of 0.338k per site at $T = 0$ K [120]. This zero-point entropy does not persist for the quantum Heisenberg model on the same lattice. However, we find again a zero-point entropy on the Kagome lattice [93].

below 1 μK , with the notable exception of helium-3 (see Problem 5.7.6). If S_0 is the entropy per unit volume due to nuclear spins, and if we do not go much below 1 mK, we can replace (1.53) with an effective statement of the third law

$$\lim_{T \rightarrow 0} \lim_{V \rightarrow \infty} \frac{1}{V} S(V) = S_0 \quad (1.54)$$

S_0 is a reference entropy that is independent of chemical composition, pressure, solid, liquid or gaseous state, crystalline form, etc. because nuclear spins are insensitive to these parameters. The third law has its origins in quantum physics, and we shall have several occasions to verify its validity in Chapter 5 from calculations in quantum statistical mechanics.

1.5.2 Application to metastable states

Consider a system that can exist in two crystalline forms (a) and (b), one stable and the other metastable but with such a long lifetime that we can apply to it the usual equations. The form (a) is stable for $T < T_c$ and form (b) for $T > T_c$ where T_c is, therefore, the temperature of the phase transition. This phase transition takes place with a latent heat L , with

$$\frac{L}{T_c} = S^{(b)}(T_c) - S^{(a)}(T_c) \quad (1.55)$$

where $S^{(a)}(T)$ and $S^{(b)}(T)$ are the entropies of (a) and (b) at T . By varying the temperature along a quasi-static path where the variables y are held constant, we can calculate $S^{(a)}(T)$ and $S^{(b)}(T)$ in terms of $C_y^{(a)}(T)$ and $C_y^{(b)}(T)$, the specific heats at fixed y defined in (1.31)

$$S^{(a)}(T_c) = S_0 + \int_0^{T_c} dT \frac{C_y^{(a)}(T)}{T} \quad (1.56)$$

and

$$S^{(b)}(T_c) = S_0 + \int_0^{T_c} dT \frac{C_y^{(b)}(T)}{T} \quad (1.57)$$

The crucial point is that S_0 , being independent of the crystalline form, is the same in the two equations. We can therefore determine the difference $S^{(b)}(T_c) - S^{(a)}(T_c)$ either by measuring the latent heat (1.55), or by measuring the specific heat and taking the difference of (1.56) and (1.57) thus eliminating S_0 . The fact that the two results coincide is a verification of the third law. A classic case is that of grey and white tin. Grey tin is a semiconductor, stable for $T < T_c = 292 \text{ K}$, white

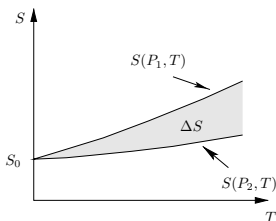


Figure 1.12 Behaviour of the entropy near $T = 0$ for two different pressures.

tin is a metal, stable for $T > T_c$. Experimental measurement of $S^{(b)}(T_c) - S^{(a)}(T_c)$ using (1.55) gives 7.7 J K^{-1} whereas the difference between (1.56) and (1.57) gives 7.2 J K^{-1} . Taking into account experimental uncertainty, the agreement is satisfactory.

1.5.3 Low temperature behaviour of specific heats

The third law constrains the behaviour of the specific heats as $T \rightarrow 0$. We start with the expression for $S(T, P)$ or $S(T, V)$

$$S(T; P, V) - S_0 = \int_0^T dT' \frac{C_{P,V}(T')}{T'} \quad (1.58)$$

and follow a quasi-static path at constant pressure or volume. Since the integral in (1.58) has to converge for $T' \rightarrow 0$ (otherwise the entropy would be infinite), C_P and C_V must vanish²³ as $T \rightarrow 0$

$$T \rightarrow 0 \quad C_P, C_V \rightarrow 0 \quad (1.59)$$

Furthermore, the expansion coefficient at constant pressure α (1.33) as well as $(\partial P / \partial T)_V$ also vanish

$$\alpha = \frac{1}{V} \left. \frac{\partial V}{\partial T} \right|_P \rightarrow 0 \quad \left. \frac{\partial P}{\partial T} \right|_V \rightarrow 0 \quad (1.60)$$

We demonstrate, as an example, the first of relations (1.60). We start with the Maxwell relation (1.28) and we plot on Figure 1.12 S as a function of T for two

²³ This argument only assumes that the entropy is finite at zero temperature.

different pressures, P_1 and P_2 . Since the two curves for S must pass through S_0 for $T = 0$, we see that $\Delta S \rightarrow 0$, while $\Delta P = P_2 - P_1$ stays finite and consequently $\Delta S / \Delta P \rightarrow 0$. A more formal demonstration goes as follows. By differentiating the Maxwell relation (1.28) with respect to T , we obtain

$$\left. \frac{\partial^2 V}{\partial T^2} \right|_P = - \frac{\partial}{\partial T} \left[\left. \frac{\partial S}{\partial P} \right|_T \right]_P = - \frac{\partial}{\partial P} \left[\left. \frac{C_P}{T} \right|_T \right]$$

On the other hand

$$\left. \frac{\partial V}{\partial T} \right|_P = - \left. \frac{\partial S}{\partial P} \right|_T = - \frac{\partial}{\partial P} \left[S_0 + \int_0^T dT' \frac{C_P(T')}{T'} \right]_T$$

The important point is that S_0 is a constant independent of P which leads to

$$\left. \frac{\partial V}{\partial T} \right|_P = - \int_0^T dT' \frac{1}{T'} \left. \frac{\partial C_P}{\partial P} \right|_T = \int_0^T dT' \left. \frac{\partial^2 V}{\partial T'^2} \right|_P = \left. \frac{\partial V}{\partial T} \right|_P - \left. \frac{\partial V}{\partial T} \right|_P^{T=0}$$

The demonstration of the result for $\partial P / \partial T_V$ follows the same lines, by using the Maxwell relation (1.27).

We can obtain an additional result for $C_P - C_V$ by making an assumption on the behaviour of C_P . We assume that C_P follows a power law in T

$$T \rightarrow 0 \quad C_P \simeq T^x f(P)$$

with $x > 0$ and $f(P) > 0$. We then obtain

$$\alpha V = \left. \frac{\partial V}{\partial T} \right|_P = - \int_0^T dT' T'^{x-1} f'(P) = - \frac{1}{x} f'(P) T^x$$

It follows that $\alpha V / C_P$ is a function only of P for $T \rightarrow 0$

$$\lim_{T \rightarrow 0} \frac{\alpha V}{C_P} = - \frac{f'(P)}{x f(P)} = g(P)$$

Using (1.38), and keeping in mind that κ_T stays finite for $T \rightarrow 0$, we see that $(C_P - C_V)$ decreases more rapidly than C_P and C_V separately

$$T \rightarrow 0 \quad \frac{C_P - C_V}{C_P} \sim \alpha T$$

1.6 Exercises

1.6.1 Massieu functions

1. Construct the three Massieu functions

$$\Phi_1\left(\frac{1}{T}, V, N\right) \quad \Phi_2\left(\frac{1}{T}, \frac{P}{T}, N\right) \quad \Phi_3\left(\frac{1}{T}, V, \frac{\mu}{T}\right)$$

2. Applications

Consider a system where the energy is a function only of the temperature T . Show that the volume is a function only of P/T , $V = V(P/T)$, and verify the immediate consequence of this (see also Problem 1.7.1)

$$\left.\frac{\partial P}{\partial T}\right|_V = \frac{P}{T} \quad (1.61)$$

Show that the difference between the specific heat at constant pressure and constant volume depends only on P/T

$$C_P - C_V = -\left(\frac{P}{T}\right)^2 V' \left(\frac{P}{T}\right)$$

1.6.2 Internal variable in equilibrium

We consider a system at fixed volume and particle number. The system is characterized either by its total energy E or its total entropy S , as well as one internal variable y . For example in the system in Figure 1.3, y may be the position of the partition, or the distribution of the energy between the two compartments. At equilibrium and for a given E , the value of y is fixed by the condition of maximum entropy

$$\left.\frac{\partial S}{\partial y}\right|_E = 0 \quad \left.\frac{\partial^2 S}{\partial y^2}\right|_E \leq 0 \quad (1.62)$$

We have seen in Section 1.4.1 that the condition of maximum entropy at fixed E becomes a condition of minimum energy at fixed entropy. We give another demonstration of this result by showing that (1.62) implies

$$\left.\frac{\partial E}{\partial y}\right|_S = 0 \quad \left.\frac{\partial^2 E}{\partial y^2}\right|_S \geq 0 \quad (1.63)$$

which is equivalent to (1.47).

1. Show first that

$$\left.\frac{\partial S}{\partial y}\right|_E = -\left.\frac{\partial E}{\partial y}\right|_S \left.\frac{\partial S}{\partial E}\right|_y \quad (1.64)$$

and then establish (1.63).

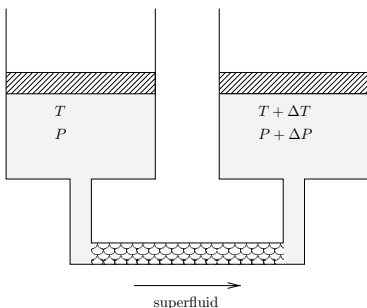


Figure 1.13 Superfluid flow through a porous plug.

2. Return to Figure 1.3 and suppose that the two compartments have different temperatures and pressures, and that the piston, while still adiabatic, is made mobile. Show that the condition of maximum entropy does not allow us to fix an equilibrium position.

3. Below 2 K, helium-4 may be considered as a mixture of a fluid of zero entropy flowing with zero viscosity through very small openings, and a normal fluid (see Problem 8.6.6 for more detail). In the setup shown in Figure 1.13, the tube connecting the two containers of fixed volumes allows only the superfluid to pass. With the help of the minimum energy condition, show that the chemical potentials are equal in the two containers, but that the pressures and temperatures may be different. If ΔP and ΔT are the pressure and temperature differences between the two containers, show that

$$\frac{\Delta P}{\Delta T} = \frac{S}{V}$$

This result illustrates the ability of the superfluid to balance a pressure difference with a temperature difference, provided the normal fluid is unable to move.

1.6.3 Relations between thermodynamic coefficients

Demonstrate the following relation

$$\frac{\kappa_S}{\kappa_T} = \frac{C_V}{C_P} \quad (1.65)$$

1.6.4 Contact between two systems

Consider two identical copper blocks, A and B , of mass m , specific heat C and respective temperatures T_A and T_B with $T_A \leq T_B$. We assume that C does not depend on the temperature and that the volumes of the blocks are constant. We put A and B in contact while maintaining the ensemble thermally insulated.

1. What is the change in the internal energy of each of the blocks? What is the total energy change? What is the final temperature?
2. What is the entropy change of each of the blocks? What is the total entropy change?
3. We construct a heat engine which uses the two blocks as heat sources. What is the maximum work that can be obtained? What is the final temperature of the two blocks in this case?

1.6.5 Stability conditions

1. Establish the following relations at constant N

$$\left. \frac{\partial \mu}{\partial V} \right|_T = \frac{V}{N} \left. \frac{\partial P}{\partial V} \right|_T$$

$$\left. \frac{\partial T}{\partial V} \right|_S = - \frac{T}{C_V} \left. \frac{\partial P}{\partial T} \right|_V$$

2. An experimentalist claims to have found a material with the following properties

$$(i) \left. \frac{\partial P}{\partial V} \right|_T < 0 \quad (ii) \left. \frac{\partial P}{\partial T} \right|_V > 0 \quad (iii) \left. \frac{\partial \mu}{\partial V} \right|_T < 0 \quad (iv) \left. \frac{\partial T}{\partial V} \right|_S > 0$$

Which of the relations above is compatible with the stability conditions? Are all these relations compatible with each other?

1.6.6 Equation of state for a fluid

A fluid has the following properties

- (i) At a given fixed temperature T_0 , the work $\mathbb{W}$ furnished to the outside world during isothermal expansion from volume V_0 to V is

$$\mathbb{W} = RT_0 \ln \frac{V}{V_0}$$

- (ii) Its entropy is given by (a is a positive constant)

$$S(T, V) = R \left(\frac{V_0}{V} \right) \left(\frac{T}{T_0} \right)^a$$

What is the equation of state of this fluid?

1.7 Problems

1.7.1 Reversible and irreversible free expansions of an ideal gas

The most general way to define an ideal gas is as follows

- (i) The internal energy E is independent of the volume at constant temperature.
- (ii) PV is a function of only the temperature: $PV = \phi(T)$.

1. Show that $\phi(T)$ is a linear function of T : $\phi(T) = aT$.
2. In what follows we consider a mole of gas and assume that the specific heat is independent of temperature

$$C_V = \frac{l}{2}R \quad l = 3, 5, \dots$$

We recall that $C_P - C_V = R$ and we write $\gamma = C_P/C_V$. Starting at an initial temperature T_0 and initial volume V_0 , calculate the entropy $S(V, T)$ as a function of R, l and $S(V_0, T_0)$. Repeat but now taking P_0 and T_0 as initial conditions.

3. Using the fact that the entropy is constant for a reversible adiabatic expansion, show that $PV^\gamma = \text{const}$ or $TP^{(1-\gamma)/\gamma} = \text{const}$.
4. We subject a mole of gas, initially in a state (P_i, T_i, V_i) , to a reversible adiabatic expansion taking it to the state (P_f, T_f, V_f) with $P_f < P_i$. Calculate T_f and show that the work done on the gas is $W = C_V(T_f - T_i)$.
5. A mole of gas is compressed by a piston on which is placed a mass m such that the initial pressure is

$$P_i = P_f + \frac{mg}{A}$$

where P_f is atmospheric pressure and A is the area of the piston (Figure 1.14). The mass is instantaneously removed, we assume the gas to be perfectly isolated thermally and the piston to be frictionless. What is the final temperature T'_f as a function of T_i , P_i/P_f and γ ? Plot representative curves for T_f/T_i and T'_f/T_i as a function of P_f/P_i for $0 \leq P_f/P_i \leq 1$.

6. Could we have predicted without calculation that $T'_f > T_f$? Which expansion, that of Question 4 or Question 5, furnishes more work to the outside world? In the experiment of Question 5 we put back the mass on the piston. The final temperature and volume are T''_f and V''_f . Compare T''_f and V''_f with T_i and V_i .

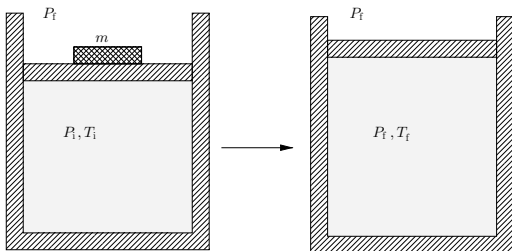


Figure 1.14 Compression and expansion of a gas.

7. In the experiment of Figure 1.8, the (ideal) gas is at a pressure P and temperature T , and the container is in vacuum. By freeing the piston, we allow the gas to expand by dV . What is the final temperature? What is the entropy increase? Deduce that although the transformation is infinitesimal it is not quasi-static.
8. In Figure 1.9, the ideal gas is initially in the left compartment, which is assumed adiabatic. A very tiny hole is made in the central partition. What is the final temperature? What is the entropy increase? Show that although this transformation is infinitely slow, it is not quasi-static.

1.7.2 van der Waals equation of state

Consider a gas whose equation of state is given by

$$\left(P + \frac{a}{v^2}\right)(v - b) = kT \quad (1.66)$$

where v is the specific volume, $v = V/N$. This is the equation of state of a ‘van der Waals gas’. It is an empirical equation that includes some of the residual interactions between molecules (attractive at long range, repulsive at short range) that were ignored in the ideal gas approximation.²⁴

1. Give a physical interpretation of the terms a/v^2 and b . In which limit does the van der Waals equation become the ideal gas equation of state?
2. Let $\check{\epsilon}(T, v)$ and $\check{s}(T, v)$ be respectively the internal energy and entropy per gas molecule. Show that the constant volume specific heat per molecule, c_v , depends

²⁴ We emphasize that the van der Waals equation is in no way an exact equation.

only on the temperature. Assume we know the internal energy and entropy of a macroscopic state at temperature T_0 and with a molar volume v_0 , calculate $\check{\epsilon}(T, v)$ and $\check{s}(T, v)$. Note: The calculations involve an integral that need not be evaluated.

3. We suppose from this equation of state that c_v does not depend on temperature. Show that

$$\begin{aligned}\check{\epsilon}(T, v) &= c_v T - \frac{a}{v} + \text{const} \\ \check{s}(T, v) &= c_v \ln T + k \ln(v - b) + \text{const}\end{aligned}\tag{1.67}$$

4. Show that for an adiabatic quasi-static process we have

$$\begin{aligned}T(v - b)^{\gamma-1} &= \text{const} \\ \left(P + \frac{a}{v^2}\right)(v - b)^{\gamma} &= \text{const}\end{aligned}$$

where $\gamma = \frac{c_v + k}{c_v}$.

5. Determine the sign of the temperature change of this gas when it undergoes an adiabatic expansion in vacuum as in Problem 1.7.1 Question 8. We assume that the initial (final) state is at T_1 and v_1 (T_2 and v_2).

6. Consider the example of Figure 1.3 and assume each compartment to contain half a mole of the van der Waals gas. Also assume that the two compartments are at the same temperature and that the initial volumes are V_1 and $2V - V_1$. What is the final temperature if the diathermal piston is free to move?

1.7.3 Equation of state for a solid

An often used phenomenological equation of state for solids is

$$E = Ae^{b(V-V_0)^2} S^{4/3} e^{S/3R} = f(V)g(S)\tag{1.68}$$

where A , b and V_0 are positive constants and R is the ideal gas constant; E is the internal energy, S the entropy and V the volume.

1. Show that this equation of state satisfies the third law of thermodynamics.
2. Show that the constant volume specific heat, C_V , is proportional to T^3 at low temperature (Debye law), and that it approaches $3R$ at high temperature (Dulong–Petit law).
3. Calculate the pressure P . What is the physical interpretation of V_0 ? How does the thermal expansion coefficient at constant pressure behave for $P = 0$? Is this behaviour reasonable?

4. Show that

$$\left. \frac{\partial S}{\partial V} \right|_T = - \frac{f'(V)g'(S)}{f(V)g''(S)}$$

and verify that the thermal expansion coefficient at constant pressure may be written as

$$\left. \frac{\partial V}{\partial T} \right|_P = C_V \frac{f'(V)g''(S)}{(f'(V))^2(g'(S))^2 - f(V)f''(V)g(S)g''(S)} \quad (1.69)$$

1.7.4 Specific heats of a rod

Consider a rod whose relaxed length is L_0 . Assume that the tension τ in this rod when stretched to a length L is given by

$$\tau = aT^2(L - L_0)$$

where L_0 and a do not depend on the temperature. We define the constant length specific heat of the rod by

$$C_L \triangleq T \left. \frac{\partial S}{\partial T} \right|_L$$

where S is its entropy. In addition we know C_L as a function of temperature when the rod is relaxed

$$C_L(L_0, T) = bT$$

where b is a constant.

1. Write the expressions for the differentials dE and dS of the rod internal energy and entropy in the variables (T, L) .
2. Calculate the specific heat $C_L(L, T)$ for arbitrary length L .
3. Let the temperature of the rod at length L_0 be T_0 . Calculate its entropy $S(L, T)$ in terms of $S(L_0, T_0)$.
4. The rod is in an initial state characterized by L_i and T_i . By applying an external force on it, its length is taken adiabatically and reversibly to L_f with $L_f > L_i$. Give the expression for the final temperature T_f and verify that the lengthening of the rod is accompanied by cooling.
5. Calculate the specific heat at constant tension $C_\tau(L, T)$. How does the rod behave when it is heated at constant tension?

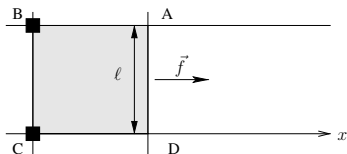


Figure 1.15 Soap film on a frame with a movable side.

1.7.5 Surface tension of a soap film

A soap film is held by the four sides of a rectangle ABCD (Figure 1.15). The wire AD, length ℓ , can be moved while kept parallel to BC allowing the film to be stretched. The modulus of the force $\vec{f}$, applied in the positive x direction, needed to keep AD fixed is $\sigma \ell$, where σ is the surface tension.

1. Write the $T dS$ equation for this problem.
2. We define the specific heat at constant length as usual

$$C_x = T \left. \frac{\partial S}{\partial T} \right|_x$$

Express the partial derivatives of the energy E in the variables (T, x) in terms of C_x , T and the partial derivatives of the surface tension σ .

3. For a wide range of temperatures around $T = 300$ K, the surface tension of soapy water varies linearly with T

$$\sigma = \sigma_0(1 - a(T - T_0)) \quad (1.70)$$

where σ_0 , a , T_0 are constants, $\sigma_0 = 8 \times 10^{-2} \text{ N m}^{-1}$, $a = 1.5 \times 10^{-3} \text{ K}^{-1}$, $T_0 = 273.16 \text{ K}$.

We stretch the film by dx *quasi-statically* and at *constant temperature*. Calculate the concomitant infinitesimal increase in the internal energy dE . Verify that energy, in the form of heat, must be given to the film in order to maintain its temperature constant.

4. Verify that when σ obeys (1.70), C_x is independent of the length of the film.
5. We may define the 'isothermal shrinkage' coefficient by

$$\kappa_T = -\frac{\ell}{\mathcal{A}} \left. \frac{\partial x}{\partial f} \right|_T$$

where $\mathcal{A}$ is the area of the film. What do you obtain for κ_T by studying the behaviour of $(\partial f / \partial x)_T$?

6. We assume that in the range of temperatures under consideration C_x remains constant. We stretch the film *quasi-statically* and *adiabatically* by dx . Calculate the resulting temperature increase dT that accompanies this stretching for an initial temperature $T = 300$ K. Assume that a reasonable order of magnitude estimate for C_x is obtained by calculating it from the bulk specific heat of the water forming the soap film. The film thickness is $e = 10^{-3}$ cm, $x = 2$ cm and we take for the bulk specific heat $C = 4185 \text{ J kg}^{-1} \text{ K}^{-1}$.

1.7.6 Joule–Thomson process

During a Joule expansion, a gas initially confined in a fixed volume escapes freely without exchanging heat with its environment. In this problem we consider a Joule–Thomson expansion where a gas initially in a compartment of volume V_1 , temperature T_1 and pressure P_1 , expands into a container of volume V_2 under a constant pressure P_2 ($P_2 < P_1$). The two compartments are thermally insulated and are connected by a porous plug which allows us to maintain the pressure difference. We assume the gas flux to be stationary. This adiabatic (but not quasi-static) expansion brings the temperature of the gas in the second compartment to a value T_2 . We shall determine the sign of the temperature change.

1. Show that this expansion takes place at constant enthalpy $\overline{H}$.
2. Can we cool down an ideal gas with the Joule–Thomson expansion?
3. The quantity $\mu_{JT} \triangleq (\partial T / \partial P)_{\overline{H}}$ is called the Joule–Thomson coefficient. Express μ_{JT} in terms of V , T , C_P and α . Verify that the value μ_{JT} takes in the case of an ideal gas is consistent with your answer to Question 2.
4. Figure 1.16 shows isenthalpy curves ($\overline{H} = \text{const}$) in the (T, P) plane for gaseous nitrogen. Comment on this figure. What does the dashed line represent? What is the maximum temperature beyond which nitrogen cannot be cooled with Joule–Thomson expansion?

1.7.7 Adiabatic demagnetization of a paramagnetic salt

When a paramagnetic solid is subjected to a uniform external magnetic field, it acquires a non-zero average macroscopic magnetization $\mathcal{M}$. An infinitesimal increase dB in the applied field B produces a change $\delta W = -\mathcal{M} dB$ in the internal

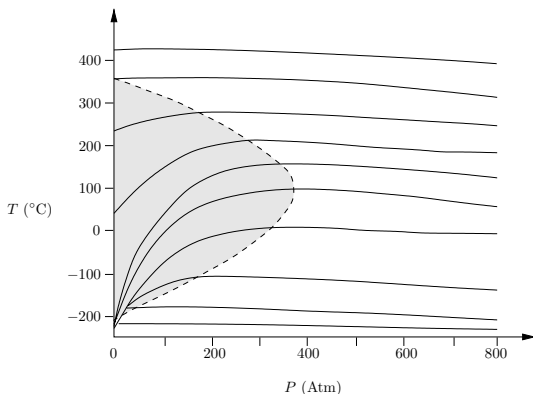


Figure 1.16 Isenthalpy curves in the (T, P) plane.

energy in the form of work. We define $\mathcal{M}$ from its relation to the magnetic susceptibility per unit volume $\chi(T, B)$

$$\chi = \frac{1}{V} \frac{\partial \mathcal{M}}{\partial B}$$

where V is the volume of the solid. We will assume that χ is given by

$$\chi = \frac{1}{V} \frac{\mathcal{M}}{B}$$

The adiabatic demagnetization of a paramagnetic crystal (also called paramagnetic salt) was one of the first methods used to attain sub-kelvin temperatures. The principle is as follows. A paramagnetic material, initially in thermal equilibrium with a heat bath at temperature T_i , is subjected to a magnetic field B_i . Although work is exchanged during this process, the heat bath maintains the crystal at its initial temperature. The solid is then thermally insulated and the magnetic field is reduced adiabatically and quasi-statically to a value B_f (usually $B_f = 0$). The final temperature is lower than the initial one.

1. Give, without calculation, the analogous mechanical equivalent (gas, piston, etc.) of this cooling process.

2. In order to characterize the temperature change accompanying the demagnetization, we introduce the coefficient $\mu_D = (\partial T / \partial B)_S$. Show that

$$\mu_D = - \frac{VTB}{C_B} \left. \frac{\partial \chi}{\partial T} \right|_B$$

where $C_B(T, B)$ is the specific heat at constant magnetic field.

3. Show that in order to calculate $C_B(T, B)$, and therefore μ_D , it is sufficient to know $\chi(T, B)$ and $C_B(T, 0)$.

4. In the temperature range (which depends on the crystal) where adiabatic demagnetization is used, we may assume that the magnetic susceptibility is given by the Curie law $\chi = a/T$ and that the heat capacity in zero field is given by $C_B(T, 0) = Vb/T^2$, where a and b are constants. Determine the final temperature T_f , after adiabatic demagnetization, in terms of T_i , B_i and B_f .

5. Calculate T_f under the following conditions: $T_i = 2$ K, $B_i = 0.71$ T, $B_f = 0$ T, $a = 78.7$ J K T⁻² and $b = 2.65$ J K.

6. By assuming the atoms in the solid to be independent, an argument based on typical energy scales in the problem leads to the conclusion that $S = S(B/T)$. With such a hypothesis, what is the minimum temperature attainable by adiabatic demagnetization? What can you conclude from this result concerning the constant b ?

1.8 Further reading

The discussion in Sections 1.1 and 1.2 is inspired by the presentation of thermodynamics in Callen's book [24] to which we refer the reader for useful comments. See in particular Chapters 1 to 8. The book by Balian [5], Chapters 5 and 6, also gives insight into this presentation. In addition there is a large number of books on thermodynamics: a classic is Zemansky [124]. Chapters 2 to 5 of Reif [109] or Chapters 1 to 5 of Schroeder [114] are strongly recommended; the latter show a great variety of applications. The reader can find discussions of long range forces in Balian [5] (Section 6.6.5), Reif [109] (Chapter 11) or Mandl [87] (Chapter 1). A demonstration of relation (1.51), which gives the curvature of a surface, is given for example in [34] Chapter 2. The main technical principles of cooling are discussed by Lounasmaa [82]; see also a more recent article by the same author [83]. The zero point entropy is discussed by Wannier [120], Mila [93] and see also Ramirez *et al.* [106].